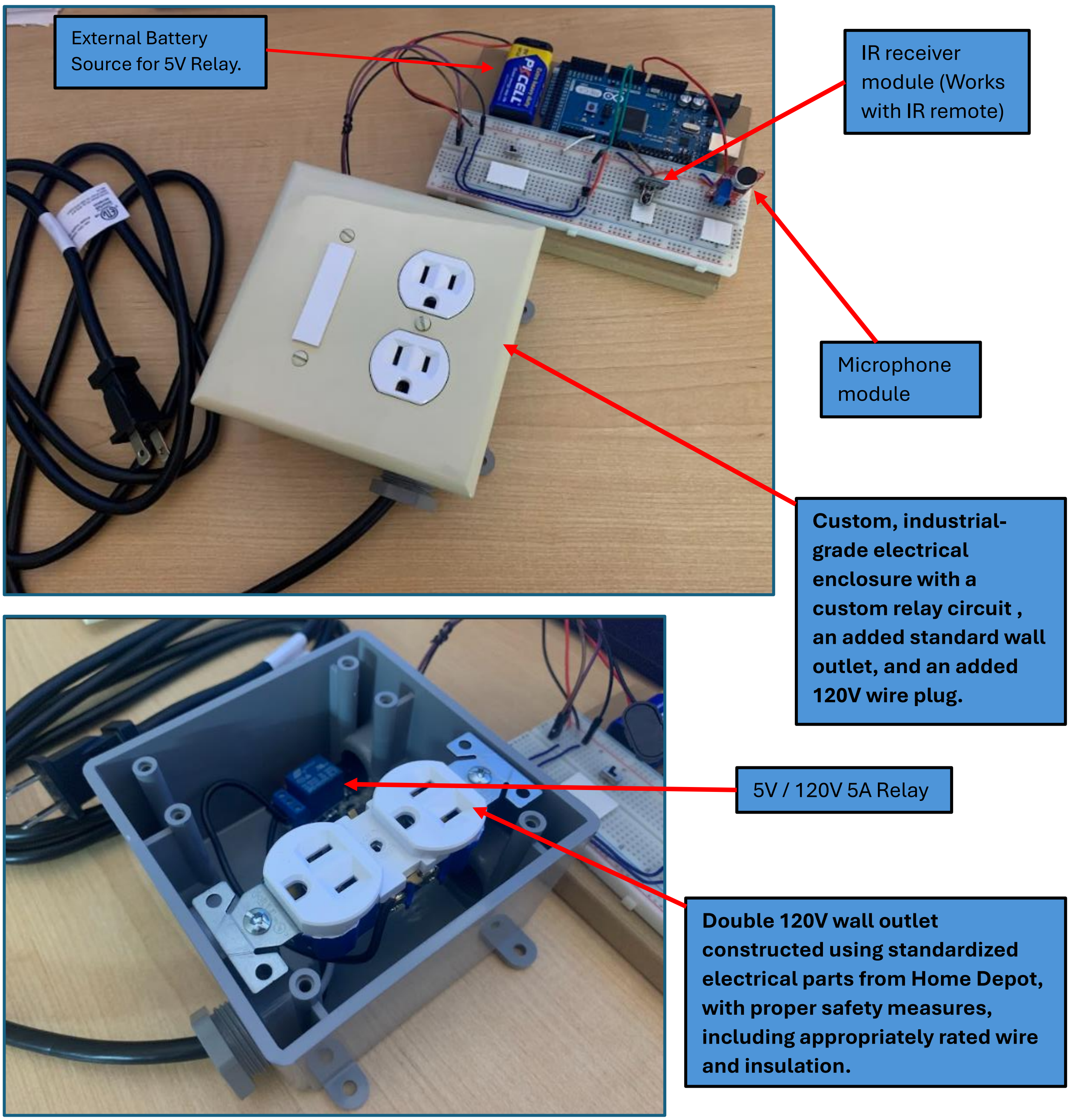
George Morales | CS-485-3 | 12/4/2024 | [GitHub Link](#)

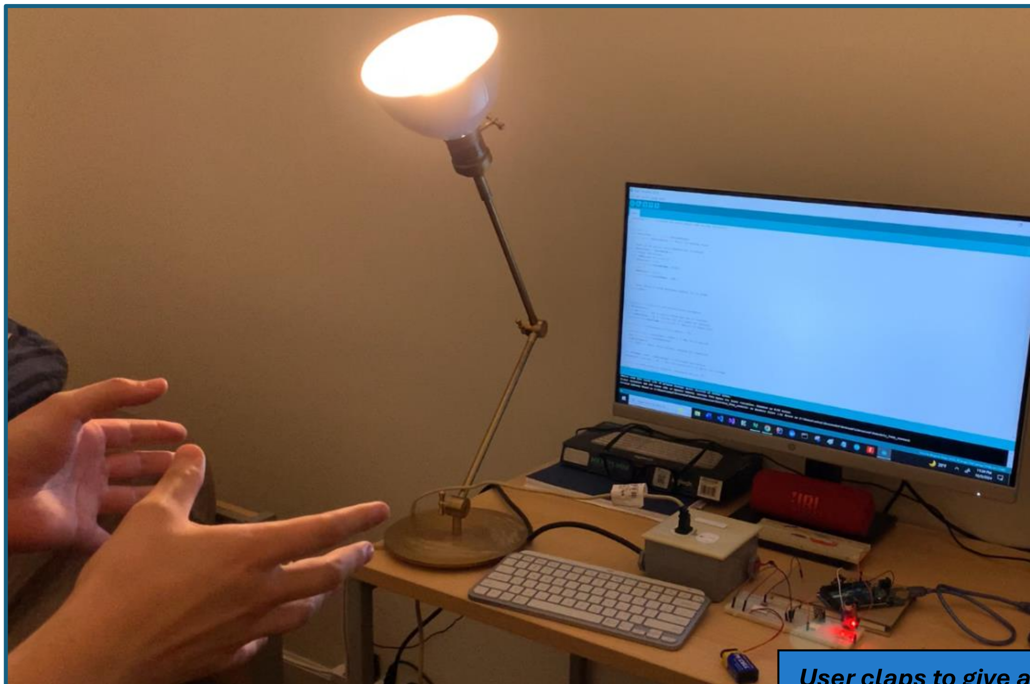
Final Project

Universal Automation Box and Design Processes ([Demos Video Link](#)):

The project involves the development of a Universal Automation Box—an Arduino-based circuit designed to bring smart automation to household appliances. At its core, this device uses an Arduino microcontroller, paired with a relay module and two sensors for sound detection and for IR remote detection to automate the operation of various home appliances like lamps and kitchen equipment.

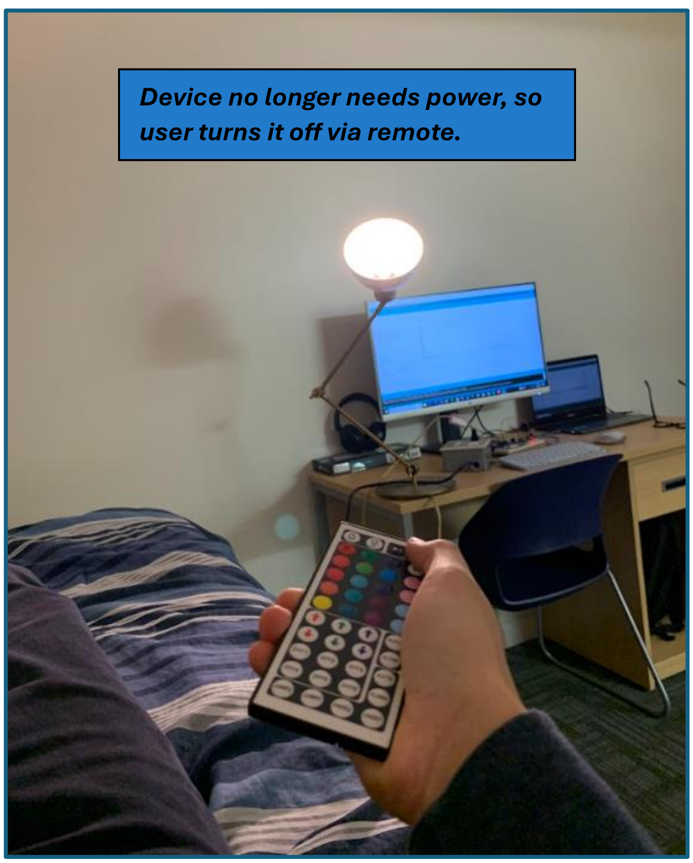


Interactions with Universal Automation Box (UAB):

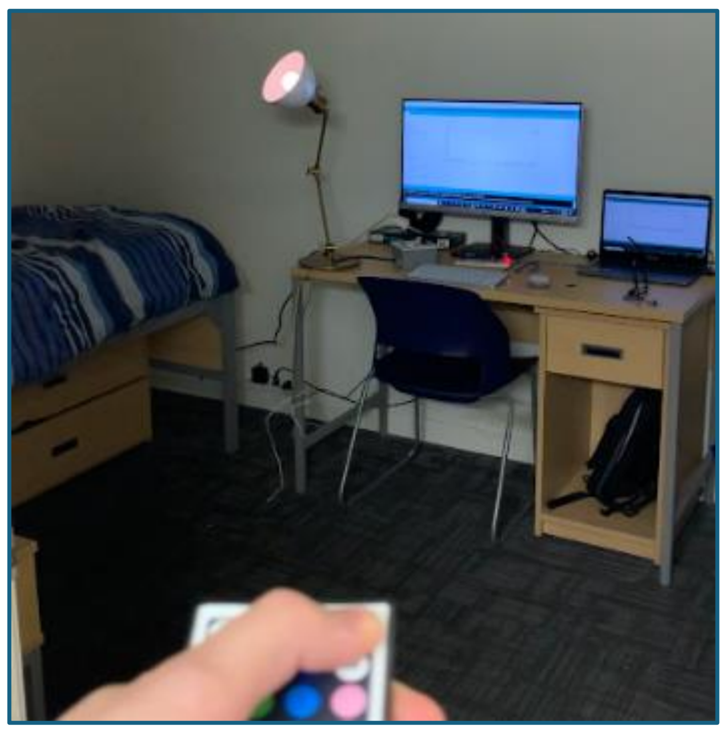


Tested interactions with household members. Initially, both sensors (IR remote and sound) were active simultaneously. However, testing revealed that the IR receiver's current draw interfered with the analog readings of the sound sensor. To address this, a switch was added to toggle between input from the sound sensor and the IR receiver module.

User claps to give appliance or device power!



Device no longer needs power, so user turns it off via remote.



User is leaving living space and can turn device off to save energy.

Elvator Pitch Drafts:

1) How can we balance automation, sustainability, and user control in modern homes?

The **Universal Automation Box** tackles this by turning everyday appliances into smart devices. Powered by Arduino and modular sensors like clap and motion detectors, it controls outlets to automate lighting, kitchen tools, and more. Focused on sustainability, it uses durable materials, supports upgrades, and works with existing systems. It's a step toward responsible automation that's efficient, modular, and eco-conscious.

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2) What if automation could be smarter, more sustainable, and adaptable?

The **Universal Automation Box** reimagines home automation with an Arduino-based system that pairs modular sensors like sound and infrared (remote) with relay-controlled outlets. It's designed to integrate with everyday appliances, making them smart while reducing waste and promoting upgradeability. By focusing on user interaction and durable materials, it's a simple, sustainable way to bring modern tech to any home.

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3) **[Final Draft]** How can we create a sustainable, modular home automation system that works seamlessly with existing appliances while promoting interaction and reducing waste?

Enter the **Universal Automation Box**—an affordable, Arduino-based solution that transforms ordinary devices into smart systems. With modular sensors like clap and infrared reciever and relay-controlled outlets, it automates appliances like lamps and kitchen tools while emphasizing sustainability. Designed for compatibility, upgradeability, and simplicity, it bridges smart tech and eco-conscious living, empowering users to automate responsibly without over-relying on technology.

5 Related Research Papers:

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1. Design and Implementation of a Wireless Home Automation System (link)

P. F. Gabriel and Z. Wang, "Design and Implementation of Home Automation system using Arduino Uno and NodeMCU ESP8266 IoT Platform," 2022 International Conference on Advanced Mechatronic Systems (ICAMechS), Toyama, Japan, 2022, pp. 161-166, doi: 10.1109/ICAMechS57222.2022.10003361. keywords: {Temperature sensors;Temperature measurement;Home appliances;Switches;Sensor systems;Sensors;Internet of Things;Internet of Things (IoT);Arduino;NodeMCU ESP8266 Wi-Fi module;Google Assistant;Blynk App},

This research outlines the design and development of a wireless home automation system using Arduino Uno and the ESP8266 NodeMCU. The system connects appliances such as lights and fans to a Wi-Fi network, enabling users to control them remotely through a mobile application. Additionally, cloud storage solutions enabled state management, ensuring devices resume their previous configurations even after power disruptions. The study emphasizes the system's sustainability potential by integrating renewable energy sources. Interestingly, the modular architecture supports scalability, allowing users to add new devices as needed. The paper concludes that such systems can enhance convenience, security, and energy efficiency in smart homes.

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2. Home Automation with Arduino and Bluetooth Integration (link)

Sharkawy, A. -N., Hasanin, M., Sharf, M., Mohamed, M., & Elsheikh, A. (2022). Development of Smart Home Applications Based on Arduino and Android Platforms: An Experimental Work. Automation, 3(4), 579-595. https://doi.org/10.3390/automation3040029

This study presents a cost-effective home automation system built with Arduino Uno and the HC-06 Bluetooth Module. The system allows mobile phone control over lighting, fans, temperature, humidity, and safety alarms for fire and toxic gases. It employs RemoteXY for a user-friendly graphical interface, achieving functionality at a low cost of $110. The study emphasizes the significance of smart home automation in improving quality of life, especially for individuals with disabilities, inpatients, and the elderly. The system essentially integrated Android mobile phones with the Arduino for seamless control of smart home features.

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3. Energy-Efficient Home Automation Using IoT Technologies (link)

S. K. Vishwakarma, P. Upadhyaya, B. Kumari and A. K. Mishra, "Smart Energy Efficient Home Automation System Using IoT," 2019 4th International Conference on Internet of Things: Smart Innovation and Usages (IoT-SIU), Ghaziabad, India, 2019, pp. 1-4, doi: 10.1109/IoT-SIU.2019.8777607. keywords: {Internet of Things;Smart homes;Home appliances;Google;Automation;Home Automation;Relay;Node MCU (ESP8266);IFTTT;Adafruit;Internet of Things (IoT);Google Assistant;Voice Control;Smartphone}, This paper proposes an energy-efficient home automation system that integrates Arduino Uno, Wi-Fi modules, and a variety of sensors. The system is designed to analyze and optimize appliance usage patterns, reducing unnecessary power consumption. Cloud platforms like Microsoft Azure are used for data storage and analytics, providing users with insights into their energy consumption. The authors emphasize the importance of real-time data logging, which enables homeowners to adjust their behaviors to save energy. By leveraging IoT technologies, this research highlights the potential for creating intelligent, sustainable, and user-centric home automation systems that benefit both users and the environment.

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4. Voice-Activated Smart Home Automation System (link)

D. -S. Lee, B. -G. Kim and S. -K. Kwon, "Efficient Depth Data Coding Method Based on Plane Modeling for Intra Prediction," in IEEE Access, vol. 9, pp. 29153-29164, 2021, doi: 10.1109/ACCESS.2021.3056687. keywords: {Predictive models;Data models;Encoding;Three-dimensional displays;Redundancy;Cameras;Video coding;Depth data;RGB-D video;video coding;video compression},

This research presents a voice-activated home automation system using Arduino, Bluetooth modules, and Google voice recognition technology. The system allows users to control basic household appliances like lights and fans through simple voice commands. A smartphone app provides additional functionality for remote control and customization of automation settings. The researchers also emphasized the accessibility and convenience of voice control, especially for the elderly and differently-abled users. Additionally, the system's modular design allows for easy expansion to include more devices. This study demonstrates how voice control can improve user experience while maintaining the affordability and simplicity of Arduino-based systems.

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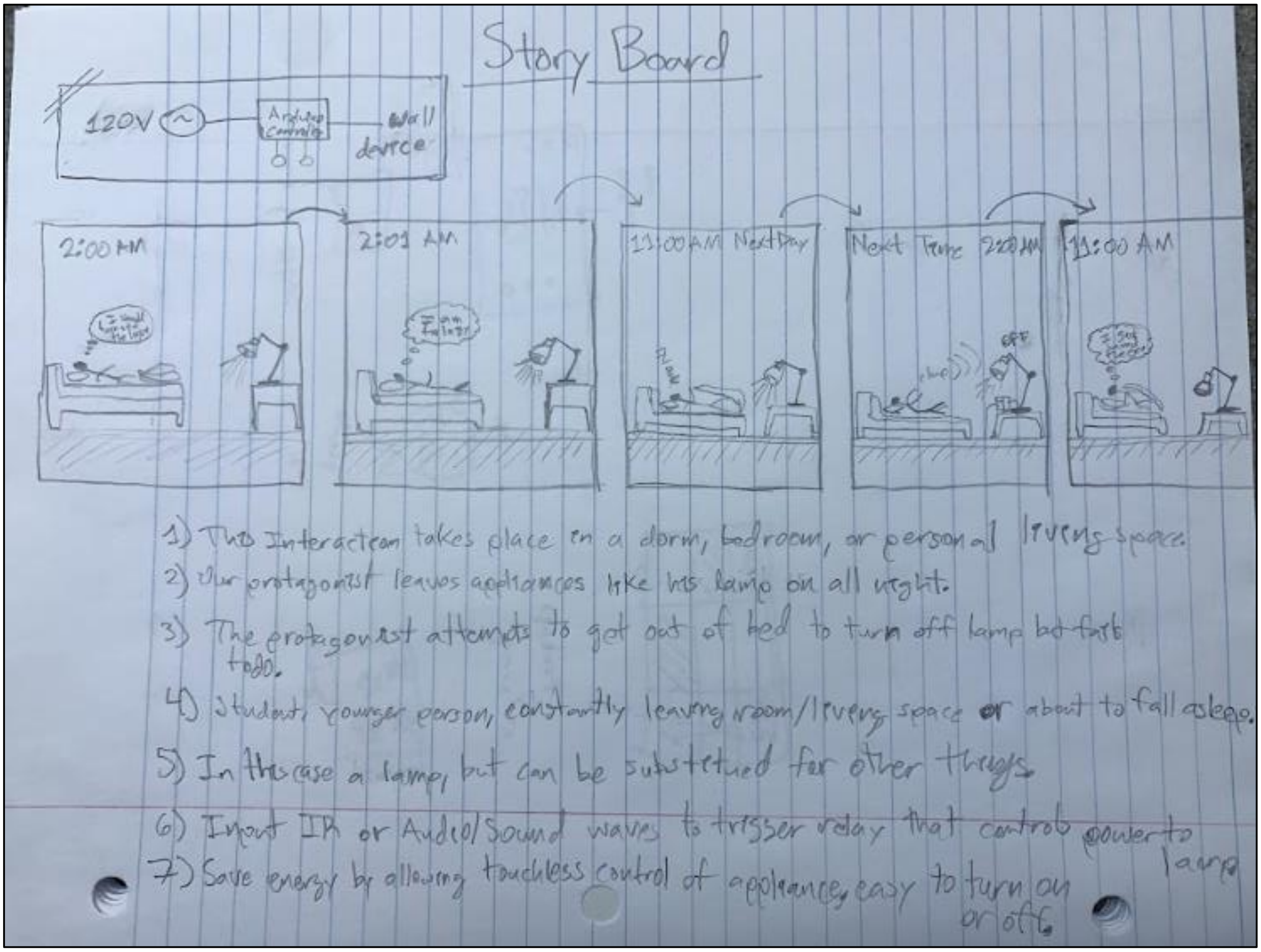
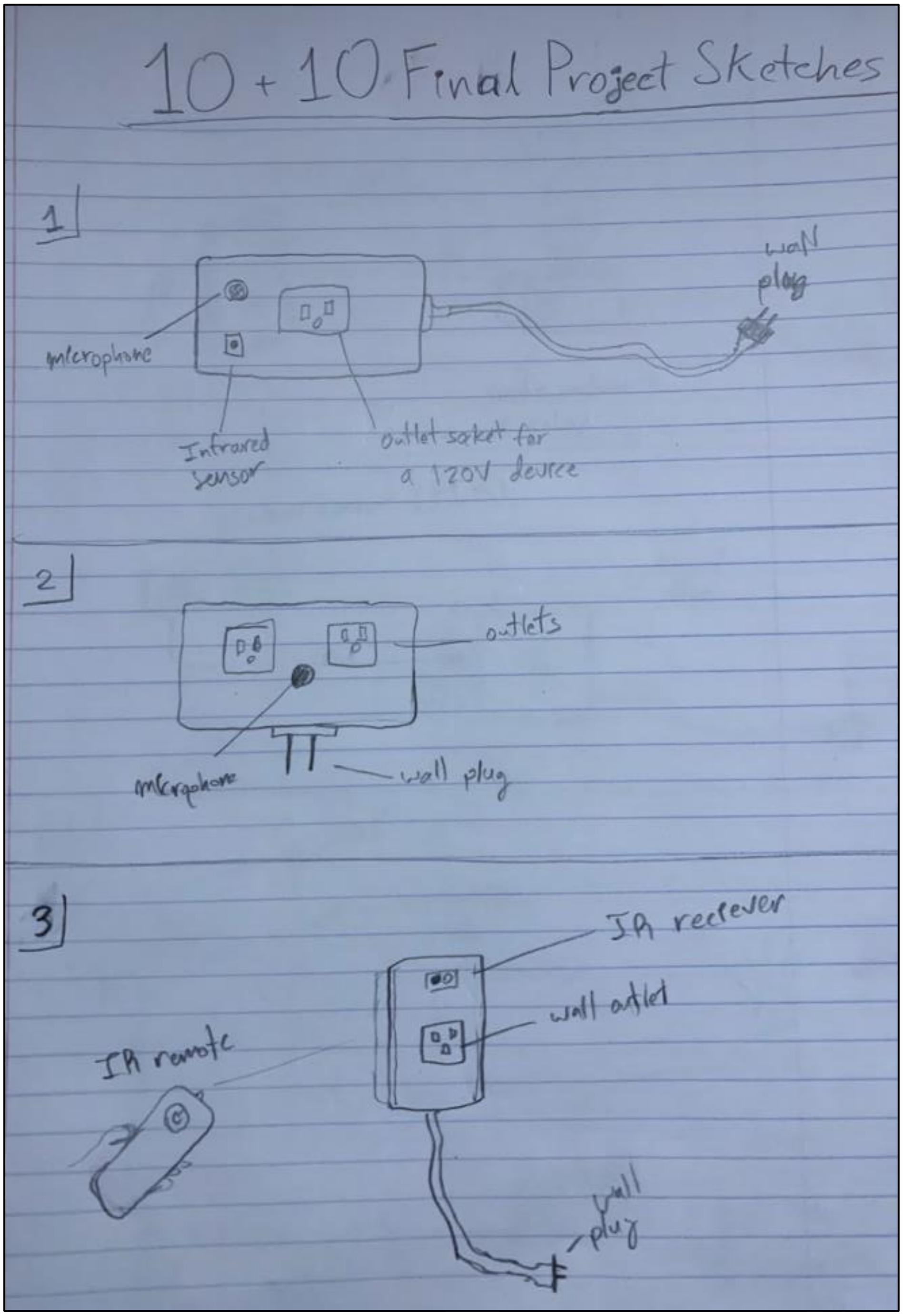
5. IoT-Based Smart Home Automation with Arduino and Sensors

This paper explores a smart home automation system that utilizes Arduino Uno as its primary microcontroller. Various sensors such as temperature, light, and motion detectors are connected alongside relays for controlling appliances. The system incorporates a smartphone app and web interface to allow real-time remote control and monitoring. The study also emphasized the affordability of the setup, which makes it ideal for low-cost implementations. The paper also highlights the energy efficiency benefits, demonstrating how IoT systems can reduce power waste and improve home energy management.

My Storyboard:



10 + 10 Sketches:



10 + 10 Sketches (Continued):